# Learning How Wrong Offline Estimates Are

## Reliable Launch Decisions for Recommender Systems

Nihar Sagar G

EECE, IIT Dharwad
PRML Course Project

10 November 2026

# The problem

- A shopping app wants to replace its recommender (the model that picks which products to show).
- Testing on real users (A/B test) is slow and risky.
- So it first **estimates offline**, from old click logs, how the new recommender would have done.
- Estimates have errors, so it launches only if

$$\text{estimated improvement} - \text{safety margin} > 0$$

## Question

Is the usual safety margin big enough? **On real data: no.** It is about half as big as it should be.

# The idea in one picture: the friend who is always late

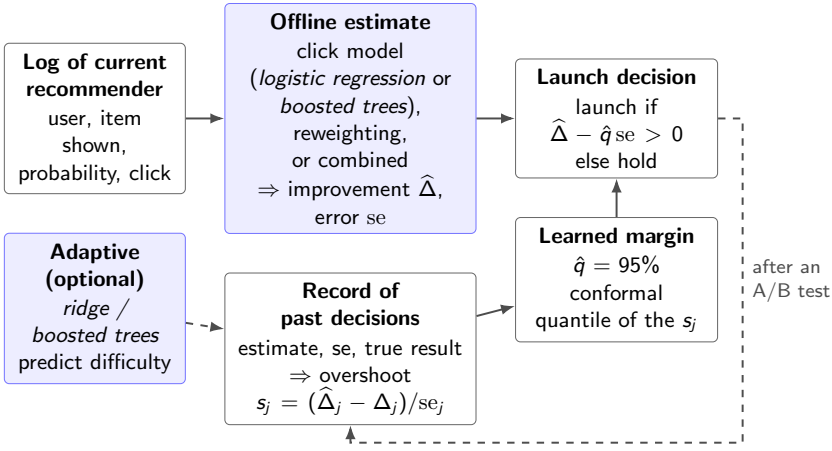
- Your friend says "10 minutes".
- Looking at the last 20 times, they arrived up to **25 minutes** late.
- So you plan for 25 minutes. You do not need to know *why* they are late.

**Our method does the same for estimates:**

- Look at past decisions where we know both the offline estimate and the real result.
- Measure how wrong the estimates actually were.
- Set the safety margin from that track record.

**Conformal prediction** is the statistically correct way to pick the number, so that it is right 95% of the time.

# The method at a glance




Blue boxes contain the machine-learning models. The dashed arrow: every A/B test adds one more past decision, so the margin keeps learning.

# Step 1: estimating a new recommender offline

Log of the current recommender: user $x$, item shown $a$, probability it was shown $p$, click $r$.

1. **Click model**: train a classifier for $P(\text{click} \mid \text{user}, \text{item})$ (**logistic regression** or **gradient-boosted trees**); average its predictions over what the new recommender would show.
2. **Reweighting**: count logged clicks, weighting items the new recommender likes more: $w = \dfrac{P_{\text{new}}(\text{item})}{P_{\text{old}}(\text{item})}$, estimate = average of $w \cdot r$.
3. **Combined**: click model, plus reweighting of its mistakes.

Improvement estimate $\widehat{\Delta}$ = new click rate − old click rate, with standard error $\mathrm{se}$.
All written from scratch; match the `obp` reference library to $10^{-9}$.

# Step 2: the learned margin

**Usual rule:** launch if $\widehat{\Delta} - 1.645\,\mathrm{se} > 0$.

**Our rule:**

1. For each past decision $j$: how many standard errors did the estimate overshoot?

$$s_j = \frac{\widehat{\Delta}_j - \Delta_j^{\text{true}}}{\mathrm{se}_j}$$

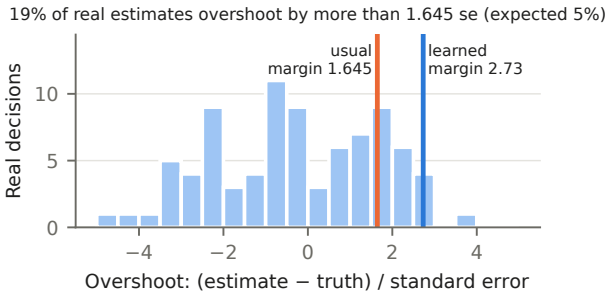
2. Sort them; take $\hat{q}$ = the value that 95% of them are below.
3. Launch if $\widehat{\Delta} - \hat{q}\,\mathrm{se} > 0$.

If estimates were honest, $\hat{q} \approx 1.645$. On real data, $\hat{q} \approx 2.7$.

## Guarantee

If the new decision is like the past ones, the margin is big enough at least 95% of the time, *even if the estimator is biased*.

# Why 1.645 is too small: the real record




- Each bar: real decisions, by how many standard errors the estimate overshot the truth.
- Usual margin assumes only 5% beyond 1.645.
- Real: **19%** beyond it.
- Learned margin **2.73**: 5% beyond.

# Why the guarantee holds

- The margin is big enough exactly when $s_{\text{new}} \leq \hat{q}$.
- If the new decision is like the past ones, $s_{\text{new}}$ is equally likely to land anywhere in the sorted list of $m + 1$ scores.
- We picked $\hat{q}$ so that at most 5% of positions are above it.
- So the margin fails at most 5% of the time.

**Assumes:** new decisions resemble past ones. **Does not assume:** unbiased estimator, correct click model, normal errors.

# Data

**Real: Open Bandit Dataset**

- 26M recommendations, fashion e-commerce site, 7 days, 3 campaigns
- Two recommenders ran *at the same time* on split traffic: one random, one adaptive
- So each one's real click rate is the **true answer** when estimating it from the other's log
- 12-hour windows $\Rightarrow$ **84 real decisions**

**Controlled: 6 classification datasets**

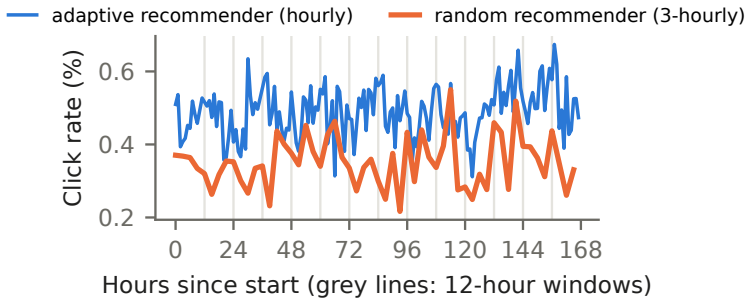
- Classes = items; click if the right class is shown
- True click rate of every recommender is exact
- New recommenders: retrained, naive Bayes, $k$-NN, random forest, ...
- Population shift created with **PCA**
- **2,688 decisions**, 27% not better

# Finding 1: real estimates are about 2× more wrong than they say

| | Actual error / reported error (1.0 = honest) | | |
|---|---|---|---|
| Estimator | usual se | grouped by page | **grouped by hour** |
| Click model (GBT) | 2.62 | 2.41 | 1.99 |
| Reweighting | 1.67 | 1.62 | 1.22 |
| Combined (GBT) | 1.80 | 1.76 | 1.13 |

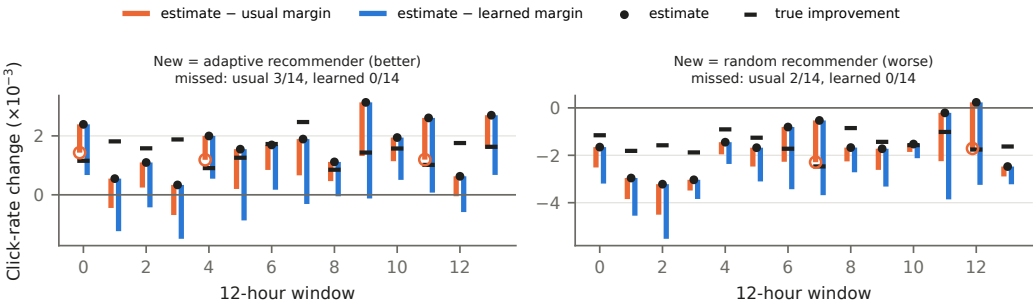
- The usual 95% margin is big enough only **60–80%** of the time.
- Grouping the 3 items on one page: no change ⇒ not the cause.
- Grouping by hour: explains most of it ⇒ **user behaviour drifts hour to hour**.
- Click models stay at ∼2: their error is **bias** from an imperfect classifier.

# Why: user behaviour drifts hour to hour



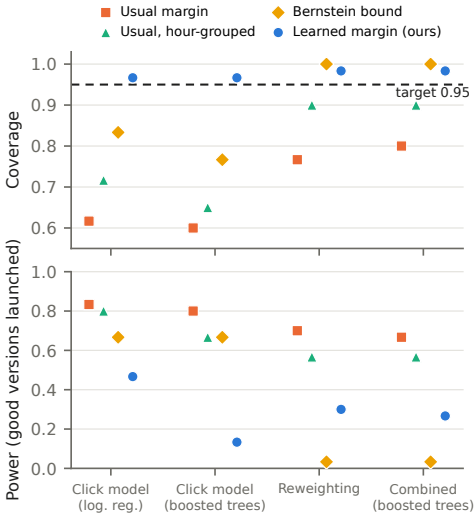

- Click rate changes within each 12-hour window (grey lines).
- The usual standard error treats every row as independent, so it misses this.
- Grouping errors by hour explains most of the gap, but not the click model's bias.

# Finding 2: every real decision, one campaign




Bars: estimate down to estimate − margin. Launch if the bar's bottom is above 0. Circle: the truth fell below the bar (a miss). All three campaigns: usual margin misses **16/84**, learned margin **4/84**.

# Finding 2: the learned margin is safe on real data

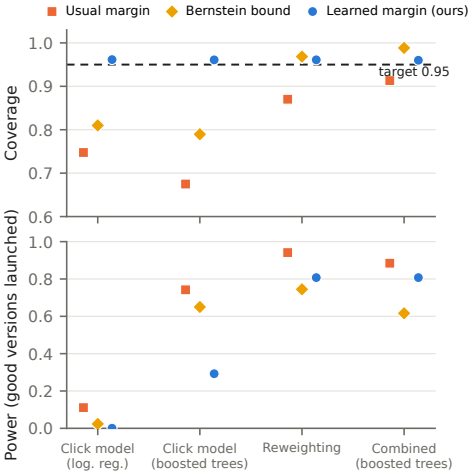



Each window uses only *earlier* windows as history.

- Usual margin: safe 60–80%
- Usual, hour-grouped: 65–90%
- **Learned margin: 95–98%**
- Bernstein bound: safe, but almost never launches

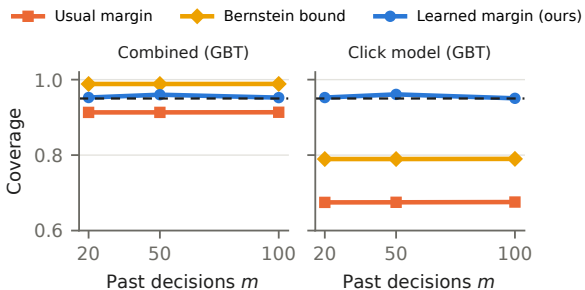
Cost: fewer launches (13–47% vs 63–83%). That is the price of honest uncertainty.

# Finding 3: controlled experiments with exact truth




- Usual margin: safe 68–91%
- **Learned margin: 95–96%** for all 10 estimator variants
- Biased click model: wrong launches **27.7%** → **0%**
- History from other datasets: still 93–95%
- Breaks under deliberate population shift (78–88%), as the assumption predicts

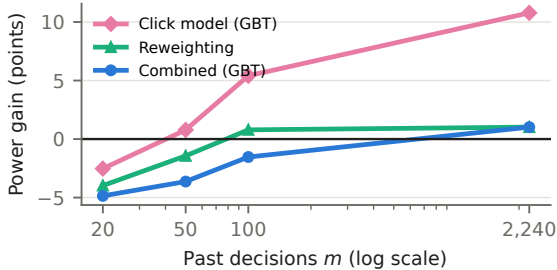
# How much history is needed?




- The learned margin hits the 95% target with only **20** past decisions.
- More history does not change coverage; it makes the margin more stable.
- The usual margin stays below target however much data you have.

# Adaptive margin: an ML model learns which decisions are hard

- Inputs: log size, largest/average weight, weight spread, $\log \mathrm{se}$
- **Ridge regression** or **gradient-boosted trees** predict how wrong the estimate will be
- Hard decisions get a bigger margin, easy ones a smaller one




**Helps only with lots of history.** With few past decisions, the plain version is better.

# Where the machine learning is

| Model | Used for |
|---|---|
| Logistic regression | Click model; the current recommender in controlled experiments; retrained versions |
| Gradient-boosted trees | Click model (stronger); difficulty model in the adaptive margin |
| Ridge regression | Difficulty model in the adaptive margin |
| Naive Bayes, $k$-NN, random forest | New recommenders in controlled experiments |
| PCA | Creates the population shift |
| Cross-fitting | Models never predict on their own training rows |
| Conformal prediction | Turns the track record into a guaranteed margin |

No deep learning. No reinforcement learning: nothing learns by trial and error; we only analyse existing logs.

# Live demo: the one window where the new recommender was worse

```
$ make demo
Estimated improvement in click rate: +0.00307 (standard error 0.00116)

Rule                                          est - margin  Decision
Usual margin (1.645 x se)                         +0.00116  LAUNCH
Usual margin, hour-grouped se                     +0.00054  LAUNCH
Bootstrap margin                                  +0.00120  LAUNCH
Bernstein bound                                   -0.00378  hold
Learned margin (2.73 x se, 72 past decisions)     -0.00010  hold

True improvement, measured on real traffic: -0.00035
  -> new recommender is NOT better
```

The usual margins launch a worse recommender. The learned margin holds.

# Conclusion and limitations

- Offline launch decisions are only as safe as their margin.
- On real logs the usual margin was about **2× too small** (hourly drift + click-model bias).
- Learning the margin from past decisions gave the promised 95% safety on real and controlled data.

**Limitations:** needs a history of past decisions; assumes the future resembles the past; fewer launches; real data has only two recommenders.

**Next:** weight past decisions by similarity; combine with hour-grouped standard errors.

# Backup: all main results

| | Coverage | | | | Wrong-launch rate | | | | Power | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Estimator | Usual | Hour | Bern. | Ours | Usual | Hour | Bern. | Ours | Usual | Hour | Bern. | Ours |
| *Real data (60 decisions, history = earlier windows)* | | | | | | | | | | | | |
| Click model (LR) | .617 | .717 | .833 | **.967** | .033 | .033 | .033 | .033 | .833 | .800 | .667 | .467 |
| Click model (GBT) | .600 | .650 | .767 | **.967** | .033 | .033 | .033 | .000 | .800 | .667 | .667 | .133 |
| Reweighting | .767 | .900 | **1.00** | **.983** | .033 | .033 | .000 | .033 | .700 | .567 | .033 | .300 |
| Combined (GBT) | .800 | .900 | **1.00** | **.983** | .033 | .033 | .000 | .000 | .667 | .567 | .033 | .267 |
| *Controlled experiments (exact truth, $m = 50$ past decisions)* | | | | | | | | | | | | |
| Click model (LR) | .748 | – | .810 | **.962** | .013 | – | .000 | .000 | .111 | – | .024 | .001 |
| Click model (GBT) | .675 | – | .790 | **.961** | .277 | – | .215 | .000 | .742 | – | .650 | .293 |
| Reweighting | .870 | – | **.968** | **.961** | .000 | – | .000 | .000 | .941 | – | .745 | .807 |
| Combined (GBT) | .913 | – | **.988** | **.960** | .005 | – | .000 | .001 | .885 | – | .616 | .808 |

Coverage: margin big enough (target 0.95, bold if met). Wrong-launch rate: not-better recommenders that were launched. Power: better recommenders that were launched. Usual = 1.645 se; Hour = hour-grouped se; Bern. = Bernstein bound; Ours = learned margin.

# Backup: adaptive margin, ridge vs boosted trees

| Estimator | Model | $m$=20 | $m$=50 | $m$=100 | Large | Real |
|---|---|---|---|---|---|---|
| Click model (GBT) | Ridge | −5.2 | −0.3 | +1.7 | +9.0 | +10.0 |
| | GBT | −2.5 | +0.8 | +5.4 | +10.8 | −6.7 |
| Reweighting | Ridge | −4.8 | +0.9 | +1.4 | +1.0 | −16.7 |
| | GBT | −4.0 | −1.4 | +0.8 | +1.0 | −13.3 |
| Combined (GBT) | Ridge | −6.8 | −1.2 | −0.6 | +0.9 | −3.3 |
| | GBT | −4.9 | −3.6 | −1.5 | +1.0 | −13.3 |

Change in power (percentage points) of the adaptive margin compared with the plain learned margin. $m$ = past decisions; Large = history from five other datasets ($m \approx 2{,}240$); Real = real-data history (24–78). Coverage stays within 0.91–1.00. Both difficulty models gain only with a large history.

# Backup: what is new compared with prior work

| Prior work | Difference |
|---|---|
| Choosing the best estimator (2023–24) | We make the *decision* safe for any estimator |
| High-confidence bounds (2015) | Valid only if the estimator is unbiased |
| Conformal intervals for single outcomes (2022) | We calibrate launch decisions using past decisions |
| Offline A/B testing (2018) | Uses only within-log variability |